# FPGA Implementation of a Railway Signaling System Using the OFDM Transmission Method

Ryo Higuchi[1], Hiroshi Mochizuki[2]
[1]Graduate School, College of Science and Technology, Nihon University
[2]Faculty, College of Science and Technology, Nihon University

Abstract: Currently, railway signaling systems that transmit control information via rails, such as the Automatic Train Control (ATC) systems in Japan, are widely used. However, increasing the transmission speed is difficult due to the transmission characteristics of the rails. To overcome this problem, we have proposed a new transmission method using Orthogonal Frequency Division Multiplexing (OFDM). We previously determined a detailed data allocation scheme using majority decision to improve the Bit Error Rate (BER) characteristics. In this paper, we describe the FPGA implementation of our proposed method and evaluate the performance of the implemented device.

## 1． Introduction

Currently, railway signaling systems using track circuits (rails) as the transmission medium, represented by systems like Automatic Train Control (ATC), are widely employed. While using track circuits as a medium offers advantages in safety, such as being less susceptible to interference compared to wireless methods, it also presents challenges. These include strong influence from unique rail noise generated by trains and the inability to achieve high transmission speeds due to the use of the audible frequency band.

To address this issue, we have been investigating a method to introduce Orthogonal Frequency Division Multiplexing (OFDM) transmission into track circuits. OFDM allows for the superposition of many carriers within a limited frequency band by narrowing the spacing between subcarriers [1]. Previous research has identified a data allocation method to improve the Bit Error Rate (BER) characteristics for realizing an OFDM-based railway signaling system [2].

Building on this, our current study focuses on the implementation of this system using a Field-Programmable Gate Array (FPGA). Specifically, we evaluate the implementation of the Fast Fourier Transform (FFT) and Inverse FFT (IFFT), which are key components of OFDM transmission. We also construct an OFDM signal generation unit and evaluate its output characteristics.

## 2． Characteristics of Transmission Using Track Circuits

Track circuits exhibit significant attenuation at high frequencies above 10 kHz. Consequently, it is not possible to use frequency bands like those in general wireless communications, restricting transmission to the audible frequency band.

Regarding noise, the system is affected by rail noise that differs from typical white noise and contains numerous harmonic components of the commercial power frequency from the return current. Figure 1 shows an example of the spectral distribution of rail noise. From this figure, it is evident that noise with peaks at multiples of 300 Hz (the 6th harmonic of the commercial power frequency) is dominant. Another characteristic is that this noise fluctuates over time.

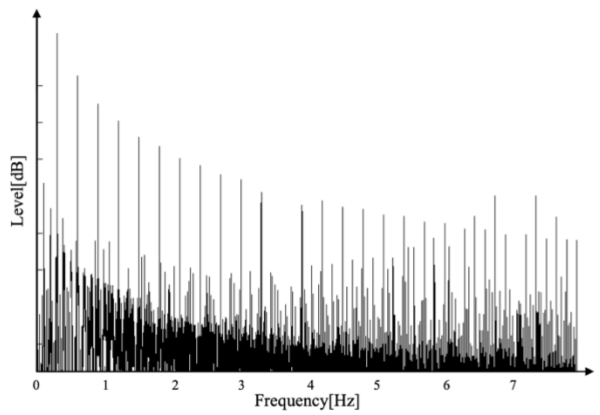


**Figure 1.** Spectral distribution of rail noise.

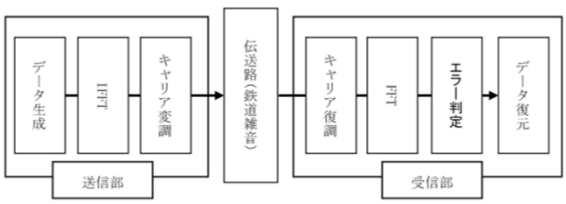


**Figure 2.** Block diagram of OFDM transmission model for railway signaling system.

## 3. Railway Signaling System Using OFDM Transmission

Based on the track circuit transmission characteristics described in the previous section, a block diagram of the proposed railway signaling system using OFDM is shown in Figure 2. OFDM is a multicarrier method that uses a large number of carriers. Even if the transmission bands of the subcarriers overlap, they do not interfere with each other due to their orthogonal relationship. This allows for high spectral efficiency, which is considered effective for narrowband transmission paths.

Furthermore, against rail noise that has peaks at specific frequencies, it is possible to improve the Bit Error Rate (BER) by allocating the same data to multiple carriers.

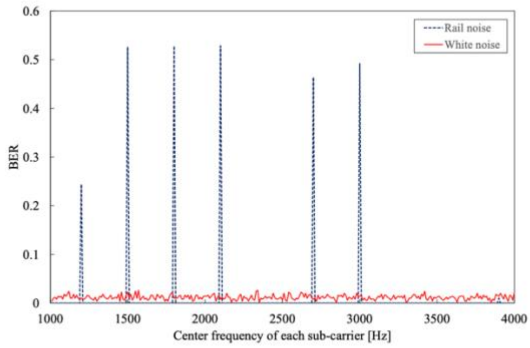


**Figure 3.** BER characteristics versus each sub-carrier.

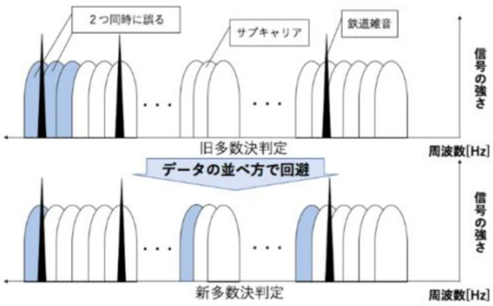


**Figure 4.** Improvement of data allocation method.

## 4. Investigation and Evaluation of FPGA Implementation

Based on our previous results, we proceeded with the actual FPGA implementation. The computer simulations in this study were conducted using MATLAB/Simulink. We leveraged this model to generate HDL (Hardware Description Language), which then served as the basis for the FPGA implementation.

Figure 5 shows a comparison of the outputs from the MATLAB/Simulink simulation model and the model implemented on the FPGA. Specifically, when data is applied to a single subcarrier, a positive output is generated in the imaginary part of the corresponding subcarrier, and a negative output is generated in the conjugate location. As Figure 5 shows, we confirmed that the FPGA-implemented model, even after modifications such as fixed-point conversion, produced an output similar to the MATLAB/Simulink model.

Furthermore, we implemented an OFDM transmitter on the FPGA that applies data to 219 subcarriers and confirmed that an OFDM signal spectrum, as shown in Figure 6, could be obtained.

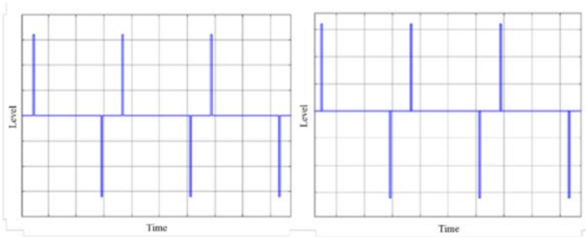


**Figure 5.** Comparison of MATLAB/Simulink-based FFT and FPGA-based FFT.

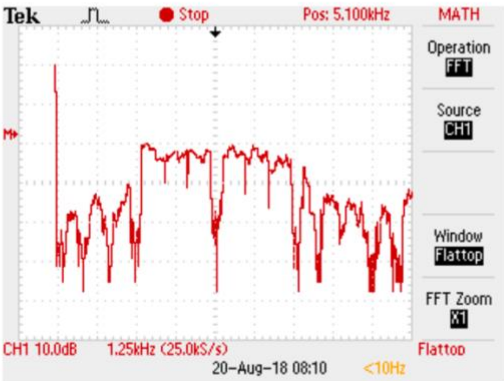


**Figure 6.** An example of spectral distribution at the transmitter.

## 5. Conclusion

In this study, we evaluated the performance of our proposed railway signaling system using OFDM transmission when implemented on an FPGA. For future work, we plan to implement the receiver unit and also conduct performance evaluations through actual field tests, with the aim of further advancing this research.

References


[1] Iwamoto et al., "A Study on the Introduction of OFDM Transmission Method to Railway Signaling Systems Using Track Circuits," 2017 National Convention of the IEEJ, 5-174 (2017).

[2] Higuchi et al., "A Study on the OFDM Transmission Method for Railway Signaling Systems," The 61st Academic Conference of the College of Science and Technology, Nihon University, G-12 (2017).